

Optical, Morphological and structural properties of Rapid Thermal Annealed CoPc thin films

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Abstract:

Effect of rapid thermal annealing temperature on structural, morphological and optical properties of cobalt phthalocyanine (CoPc) thin films is investigated. Optical absorption spectra of CoPc thin films have been employed using UV-VIS measurements performed in the wavelength range from 300 nm to 900 nm. Electronic transitions of this absorption spectrum clearly reveal Q band transitions in CoPc thin films for as-deposited and annealed samples. These Q band transition at 618 nm, 690 nm absorption wavelength suggests π - π^* transitions with an optical bandgap of 3.8 eV, 3.82 eV, 3.79 eV, 3.79 eV and 3.81 eV for as-deposited, 100 °C, 200 °C, 300 °C and 400 °C annealed samples respectively. Hall measurements of as-deposited CoPc thin films reveals the conductivity of the films as p-type, with mobility $130.05 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$ and film resistivity of $269.7 \text{ } \Omega \text{ cm}$. Raman measurements at 785nm excitation wavelength were carried out for the as-deposited CoPc thin films and found significant molecular vibrations at 1539 cm^{-1} , 755 cm^{-1} and 685 cm^{-1} which corresponds to C-C and C-N bands. The surface morphology of CoPc thin films using AFM measurements shows slight increase in the surface roughness with the increase in annealing temperature which might be due to the aggregation of native grains. The FESEM measurements with EDS spectrum of as-deposited, 100 °C, 200 °C, 300 °C and 400 °C annealed samples shows improved surface morphology through the post-deposition annealing. Highly uniform nano-spherical and isotropic grains have been observed in thin films at 400 °C annealed temperature. This physical property of CoPc films shows significant influence on the electrical and optoelectronic properties of CoPc/n-Ge heterojunction devices.

Keywords: Cobalt phthalocyanine (CoPc), Rapid Thermal Annealing, Surface morphology, Optical properties, Hall measurements and Raman effect